

APPROPRIATE HEDGING

PCA not only breaks down the driving forces of a trade (like a 5Y-10Y steepening position) into uncorrelated factors but also quantifies their impact and thereby allows us to hedge against specific factors. In order to create a 5Y-10Y steepening position which is not affected by the first factor (i.e. by direction and by directional impacts on the slope), we can simply see how changes in the first factor impact 5Y and 10Y yields and choose the hedge ratios in such a way that both net out. This leaves a position, which is hedged against changes in the first factor, that is, a steepening trade which is only affected by the non-directional steepness (and higher factors like curvature), not by directional impacts on the curve and thus a source of P&L uncorrelated to market direction (alpha).

In the PCA framework, hedge ratios are calculated in order to immunize a portfolio against changes in factors. In order to execute the non-directional 5Y-10Y steepening position of the example above, we need to hedge against changes in the directional factor α_1 , and the ratio of notionals for 5Y and 10Y is thus:

$$\frac{n_5}{n_{10}} = \frac{BPV_{10}}{BPV_5} \cdot \frac{e_{110}}{e_{15}}.$$

In the formula above, the ratio $\frac{e_{110}}{e_{15}}$ represents the quotient of 10Y and 5Y sensitivities to changes in the first factor. This may sound similar to the “beta-adjustment” of basis point value (BPV) hedge ratios by the slope of the regression line of a regression between the two instruments involved. However, as a regression minimizes the *conditional* expected value of deviations, it is conceptually quite different from a PCA. This is reflected in the practical problem that, unless the correlation is 1 or -1 , the hedge ratio determined by beta adjustment changes when the dependent and independent variable in the regression are exchanged. PCA works without conditional expected values and is hence free from this problem.

Hedge ratios against more factors are best calculated via matrix inversion. For example, the hedge ratio for a 2Y-5Y-10Y butterfly which is neutral to changes in the first and second factor can be calculated for a given notional n_5 for 5Y by:

$$\begin{pmatrix} n_2 \\ n_{10} \end{pmatrix} = \begin{pmatrix} BPV_2 \cdot e_{12} & BPV_{10} \cdot e_{110} \\ BPV_2 \cdot e_{22} & BPV_{10} \cdot e_{210} \end{pmatrix}^{-1} \cdot \begin{pmatrix} -n_5 \cdot BPV_5 \cdot e_{15} \\ -n_5 \cdot BPV_5 \cdot e_{25} \end{pmatrix}$$

We argue that hedge ratios should be based on PCA rather than on BPV neutrality. BPV neutrality corresponds to assuming arbitrarily that all entries in the first eigenvector are the same (which is in fact our very crude example

for a factor model of the first category at the beginning of this chapter). The market tells us that they are not. In other words, directional shifts do affect different points on the yield curve in different ways. If the 5Y Bund yield increases by one basis point (bp), the 10Y yield is expected to increase by 0.87 bp. True neutrality with respect to the level of yields needs to take these impacts of the direction on the curve shape into account by using hedge ratios based on PCA eigenvectors. By contrast, BPV neutrality results in positions with directional exposure.

To illustrate our point, we have regressed a BPV-neutral weighted 2Y-5Y-7Y butterfly and a PCA-neutral weighted 2Y-5Y-7Y butterfly on the Bund curve against the first factor (representing directionality), the second factor (representing non-directional steepness), and the third factor (representing curvature net of the impacts of the first and second factors on curvature, i.e. net of the impacts of direction and of non-directional steepness on curvature¹³). The results are depicted in Figure 3.14. It turns out that the BPV-neutral butterfly is not only directional but also provides little exposure to the third factor. Hence, by choosing BPV-neutral weights for a 2Y-5Y-7Y butterfly, the investor ends up with a fuzzy exposure to direction, non-directional steepness, and just a bit of net curvature. Given the strength of directional impacts, virtually all of his performance will be driven by the market direction, thus transacting Bund futures would have resulted in almost the same exposure (with fewer costs and a cleaner understanding of the risk profile). Presumably, he wanted to be exposed to factor 3 rather than a fuzzy combination of risks; however, by BPV-neutral weighting, he lost almost all net curvature exposure and ended up with a risk profile he probably did not even know of (because PCA is needed for the decomposition into factors) and is now heavily dependent on the market direction against which he thinks he is “hedged.” Note that this example is still benign in the sense that at least the correlation of the BPV-neutral butterfly to the third factor is positive. However, this does not need to be the case and there are instances where BPV-neutral steepeners are actually non-directional *flatteners*. Thus, BPV-neutral “hedging” may give any exposure, by chance the right one, a completely different one (like in the butterfly example above, which is mainly dependent on factor 1 rather than 3), or even an exposure which is opposite to the intended one. Only PCA allows the decomposition of the exposure into its components – and hence also the calculation of the “right” hedge ratio (i.e. the one leaving only the desired exposure).

¹³These verbal monstrosities are the translation of mathematical factor decomposition into everyday terms.

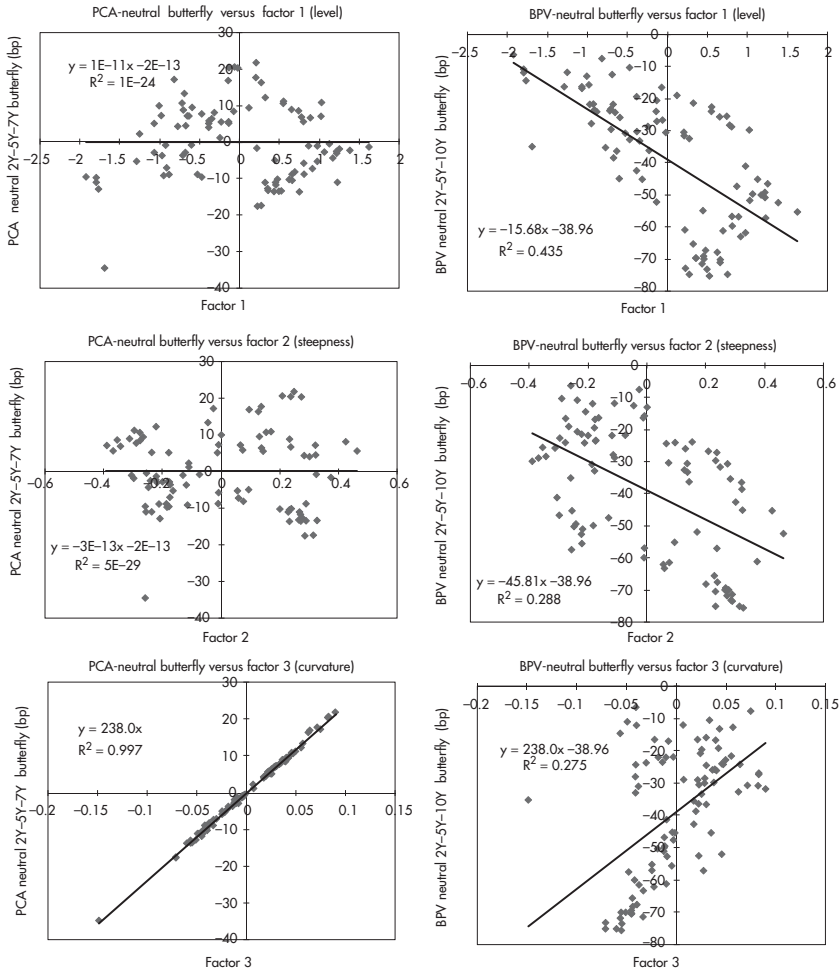


FIGURE 3.14 Driving forces of a BPV-neutral and a PCA-neutral 2Y-5Y-7Y butterfly on the Bund curve.

Sources: data – Bloomberg; chart – Authors.

Data period: 4 Jan 2010 to 3 Oct 2011, weekly data.

This is depicted in Figure 3.14 in the regression charts of the PCA-neutral butterfly, which is fully exposed to the net curvature (factor 3). Further, this is the only factor to which the PCA-neutral butterfly is exposed.¹⁴ Hence, only the PCA hedge produces a position with a clean and clear exposure to precisely those factors, to which an exposure is intended.

Note the difference of goals in PCA-neutral hedging and hedge ratios, which are calculated in order to maximize Sharpe ratios. While hedge ratios determined by maximizing Sharpe ratios (or another desired feature) obviously have the best statistical properties, PCA-neutral positions give a clear picture, to which factors a trade is exposed. And since these factors often have a meaningful economic interpretation, the investor can assess and adjust his risk profile in economic terms. For example, he knows that the PCA-neutral butterfly is hedged against directional impacts on the yield curve. One cannot say that statistically optimized hedge ratios or PCA-neutral hedge ratios are “better”, they just achieve different goals by construction: statistical optimality of a position or a position with a clear exposure to economically interpretable variables. In a manner of speaking, the investor choosing PCA-neutral hedge ratios rather than those optimizing Sharpe ratios gives up some statistical benefits in order to gain the confidence of knowing the exposure of his trade in economic terms, for example, the confidence of knowing that it is immune from directional impacts.

ANALYZING THE EXPOSURE OF TRADING POSITIONS AND INVESTMENT PORTFOLIOS

Just as in the example of Figure 3.14, any position can be regressed against the uncorrelated factors of a PCA. Thereby, its driving forces reveal themselves. This gives a clear picture of what really impacts the P&L of complex trades. For those investors using BPV-neutral hedging, the result can be quite surprising, as in the example above. Fortunately, if the actual exposure turns out to be different from the desired one, applying the appropriate PCA hedge ratios solves the problem.

For example, if a market maker has just transacted a BPV-neutral 2Y-5Y-7Y butterfly with an investor, he could run the regressions from above,

¹⁴In fact, it is also exposed to factors 4 and higher, but they have much less impact on the overall yield curve variation (and butterfly trade performance) than factor 3 does. The crucial point is to hedge against all factors of lower order than the one to which exposure is wanted.

find that he is now exposed 44% to factor 1, 29% to factor 2, and 28% to factor 3, see how this nets out with other positions in his portfolio and initiate the appropriate hedging (against those factors to which he does not want to be exposed).

Moreover, basing analysis on orthogonal eigenvectors “orthogonalizes” the mind of analysts. It helps us provide a clean and clear analysis. Which trade is exposed to which factors? Do I like that exposure or not? If not, how can I hedge against it? Do I recommend a butterfly due to its directionality or due to its curvature exposure? This breakdown of any trade into its uncorrelated driving forces prevents the all-too-common mix-up of arguments, for example, an analyst arguing via curvature for a butterfly while it has, in fact, a purely directional exposure.

Market Reconstruction and Forecasting

Reversing the decomposition of a complex market into its uncorrelated components, factor models in general allow an approximate reconstruction of the larger market from a limited number of factors.

And among all factor models, PCA does the best job in that it provides the closest possible reconstruction to reality, given limited information. This is the basis for a variety of applications, a couple of which we briefly highlight now.

Often, market participants are interested in knowing how a certain event would impact the overall market. For example, how would the yield curve be expected to react if CPI were to increase by 1%? Or what would it mean for the yield curve, if 2Y Bund yields increased by 25 bp? In mathematical terms, this corresponds to reconstructing the whole set of information (yield curve) from one single piece of information (short rates go up by 25 bp). The first eigenvector contains just this transformation, as it translates the movement of one point on the yield curve into the move of the whole curve. Using the sensitivities of Figure 3.5, an increase of 2Y yields by 25 bp would be associated with 5Y yields rising by 33 bp, 7Y by 24 bp, and 10Y by 23 bp on average. These numbers are given by multiplying 25 bp with the ratio of sensitivities to the first factor, just as in the calculation of PCA hedge ratios. Since by construction, the first factor contains maximum information, the PCA reconstruction minimizes the residuals (i.e. provides the best possible picture of the overall market, given the informational constraints).

If the reaction of the yield curve to external driving forces should be estimated, the change of an external variable can in a first step be translated via a regression into changes of factors and then the yield curve can be reconstructed from those factors. For example, a change in the EUR FX rate affects all three factors of a PCA on the Bund curve (see regression Table 3.1). Thus, the impact of a one-point change in the EUR FX rate on the whole Bund yield

curve can be assessed by calculating its impact on the first three factors (via a regression) and then reconstructing the whole yield curve from those factors.

A similar method can be used to hedge and price trading books. Imagine that since the last close, Schatz, Bobl, and Bund futures have moved by 2, 3, and 2 bp respectively. Now, a client asks a trader to buy an illiquid 7Y Bund, in which no market action has occurred since the last close, from him. The best hedge the trader can achieve in that situation is to reconstruct the yield curve (or only the 7Y point) from the information about the 2Y, 5Y, and 10Y points via the first three factors of a PCA and hedge his position in an illiquid 7Y Bund via a combination of Schatz, Bobl, and Bund futures. The hedge ratios are calculated as described above in order to achieve neutrality against the first three factors. Again, given the constraints (just three liquid instruments), this is the best hedge possible (i.e. it has smallest residual to the actual moves of the 7Y Bund).

A Yield Curve Model Based on PCA

PCA is a deterministic linear algebra tool that simply transforms the basis of the covariance matrix without introducing any stochastic process. Correspondingly, if there are n input variables, PCA will return n factors ($k = n$), and the PCA model outlined above has no residual.

One can now *artificially* introduce a residual by redefining factors as residuals. With

$$\begin{pmatrix} \varepsilon_1^t \\ \vdots \\ \varepsilon_n^t \end{pmatrix} := \sum_{i=k+1}^n \alpha_i^t \cdot \begin{pmatrix} e_{i1} \\ \vdots \\ e_{in} \end{pmatrix},$$

the PCA model

$$\begin{pmatrix} y_1^t \\ \vdots \\ y_n^t \end{pmatrix} = \sum_{i=1}^n \alpha_i^t \cdot \begin{pmatrix} e_{i1} \\ \vdots \\ e_{in} \end{pmatrix}$$

becomes

$$\begin{pmatrix} y_1^t \\ \vdots \\ y_n^t \end{pmatrix} = \sum_{i=1}^k \alpha_i^t \cdot \begin{pmatrix} e_{i1} \\ \vdots \\ e_{in} \end{pmatrix} + \begin{pmatrix} \varepsilon_1^t \\ \vdots \\ \varepsilon_n^t \end{pmatrix}$$

and now *looks like* a k -factor stochastic yield curve model.

This specific approach has two important consequences. First, the analyst can decide the number of factors to use for a PCA-based model. There is

no argument that forces him to limit the factor decomposition to two, three, or seven factors. Hence, he can choose the number of factors freely in his model. He will typically base his choice on the goal of the analysis and the structure of the eigenvalues. In particular, external knowledge about market mechanisms can be introduced via the selection of the number of factors. If, for example, in the process of eigenvector interpretation it turns out that the yield curve is driven by the first three factors, then the analyst can conduct a three-factor decomposition and justifiably treat the remaining factors as stochastic residuals.

Second, and key for the following, the residuals will continue to exhibit a factor structure when the number of factors used in the model is lower than the number of factors exhibited by the data. In the example above, the two-factor residuals will consist of both the third factor and the three-factor residuals (which, in this example, are assumed to have no clear factor structure and are therefore indiscriminately treated as noise). In general, if the eigenvalues are decreasing quickly, it will hold true that

$$\begin{pmatrix} \varepsilon_1^t \\ \vdots \\ \varepsilon_n^t \end{pmatrix} \approx \alpha_{k+1}^t \cdot \begin{pmatrix} e_{k+11} \\ \vdots \\ e_{k+1n} \end{pmatrix},$$

that is:

k -factor residual $\approx (k + 1)$ -th factor $\times (k + 1)$ -th factor loading.

In particular, the shape of k -factor residuals will correspond to the shape of the $(k+1)$ -th factor loading and the size of k -factor residuals to the size of the $(k+1)$ -th factor. Thus, k -factor residuals are high if and only if factor $(k+1)$ is high (in absolute terms). And as high residuals indicate candidates for trading opportunities, the analyst only needs to follow a few factors in order to keep track of all potential trades. This obviously greatly simplifies the task of screening the market for trade ideas.

PCA AS A TOOL FOR SCREENING THE MARKET FOR TRADE IDEAS

This factor structure means in practice that by following a few uncorrelated factors the analyst can monitor all relevant market developments in an easy and orderly manner. Outliers in the factors can be directly translated into candidates for trade ideas. By contrast, the set of all BPV-neutral butterflies,

which may at first glance appear to offer more independent trading possibilities (through different combinations, like 1-2-3, 1-2-5, 2-5-7), do not contain *more* information than the three factors (and their residuals) but just represent the same information in different and meaningless *mixtures*. For example, a 1-3-5 butterfly may mix 70% of factor 1, 25% of factor 2, and 5% of factor 3. A 2-7-10 butterfly may mix 80% of factor 1, 10% of factor 2, and 10% of factor 3. All these mixtures are endless repetitions of the same information, which can be identified and expressed cleanly, clearly, and easily through a few uncorrelated PCA factors.

Therefore, the yield curve model based on PCA can be used to screen the market for statistically attractive trading opportunities.

Applying this theory to the example of the Bund yield curve, we see with one glance on Figure 3.13 that factor 2 is close to its mean and hence it makes no sense to look for non-directional steepness positions.¹⁵ Factor 3, on the other hand, seems to be significantly away from its mean (and, unlike factor 1, exhibits a sufficient speed of mean reversion). Thus, we would focus our attention on butterfly positions (neutral against factors 1 and 2).

Note again that almost all the relevant information needed to assess the statistical features and attractiveness of yield curve trades is contained in the single Figure 3.13. A screenshot like this can therefore greatly simplify the efforts of an analyst and direct them toward the most promising targets. This is a direct exploitation of the mathematics of a PCA, which reduce almost all the information contained in the Bund curve to three numbers (factors). Correspondingly, the single Figure 3.13 contains all information about the historical evolution of the Bund market, in an orderly and clear way. This is the basis that enables in practical terms an easy and systematical screening of the whole Bund market for trades, missing none and counting no one twice. After getting used to it, one does not like to follow a market in a different form than a factor decomposition like Figure 3.13.

PCA as a Tool for Asset Selection

Having identified butterflies as candidates for curve trades, we can select the best maturities (from a statistical point of view) by looking at their two-factor residuals (i.e. the sum of the third factor times the third factor loadings and the three-factor residuals). In case of the eigenvalues decreasing quickly, this will almost equal the *shape* of the third factor loadings, and indeed the shape seen in Figure 3.15 is very similar to that of the third-factor loadings depicted in

¹⁵This discussion is obviously restricted to statistical reasons for a trade. Of course, there can be other reasons, like fundamental views or anticipated flows, for or against a trade.

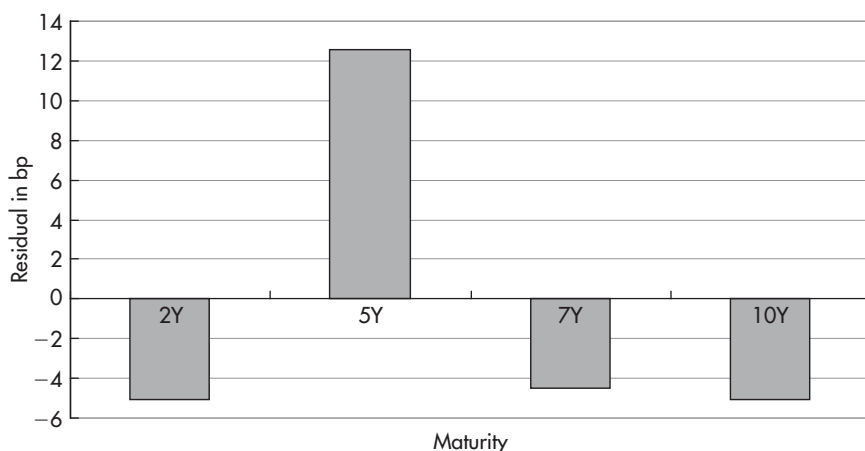


FIGURE 3.15 Two-factor residuals.

Sources: data – Bloomberg; chart – Authors.

Data period: 4 Jan 2010 to 3 Oct 2011, weekly data; current residuals as of 3 Oct 2011.

Figure 3.5 (mirrored by the x-axis since the third factor is currently negative). Consequently, a 2Y-5Y-10Y butterfly maximizes the potential profit from the current deviation of factor 3 from its mean.

Moreover, the third factor mainly determines the *size* of the two-factor residuals (i.e. the potential profit from mean reversion). Given that factor 3 is currently quite exceptionally far away from its mean (Figure 3.13), the residuals (Figure 3.15) are large by historical standards and hence the 2Y-5Y-10Y butterfly looks attractive also from a historical point of view. Note how favorably such a PCA-based top-down analysis compares to the bottom-up approach of screening all possible butterfly combinations.¹⁶

When analyzing swap curves, the selection of the best maturities can of course also be done by displaying a residual chart like Figure 3.15 for the swap rates of each maturity. Additionally or alternatively, one can use the consecutive one-year forward swap rates (i.e. the 1Y, 1Y forward 1Y, 2Y forward 1Y, 3Y forward 1Y, etc. swap rate) as input in a PCA. This is like running the analysis on the consecutive building blocks of the yield curve rather than on the

¹⁶ Actually, PCA was first applied in engineering with the aim of finding the common behavior of a production series and to detect outliers, quite similar to our application in finance, where common behavior = market mechanisms and outliers = candidates for trading opportunities.

combinations of those building blocks into usual swap rates. Since there is less information overlap between the consecutive one-year forward swap rates than between usual swap rates (the 29Y and 30Y swap rates contain the same consecutive 29 one-year forward swap rates and differ just in the last one), usually the correlation between the consecutive one-year forward swap rates is lower, which may result in better statistical properties of the PCA. Furthermore, the output of that PCA will be a residual chart showing richness and cheapness of the individual building blocks of the yield curve. For the example of a factor 3 trade on the consecutive one-year forward swap rate curve, the picture of two-factor residuals could look like the one shown in Figure 3.16.

Then, the areas of richness and cheapness can be combined. In the example of Figure 3.16, the resulting butterfly trade would be pay 3Y (the combination of 1Y, 1Y forward 1Y, and 2Y forward 1Y), receive 3Y forward 4Y (the combination of 3Y forward 1Y, 4Y forward 1Y, 5Y forward 1Y, and 6Y forward 1Y), pay 7Y forward 3Y (the combination of 7Y forward 1Y, 8Y forward 1Y, and 9Y forward 1Y). Note that this approach allows a much sharper selection of rich and cheap areas than the usual swap rates. For example, the cheap 5Y swap rate contains the rich 2Y swap rate, while the 3Y forward 4Y rate does not. Thus, combining the building blocks of the rich and cheap rate curve areas together offers the maximal exploitation of the residuals. On the other hand, transaction costs are usually higher for awkward forward rate combinations than for plain vanilla swap rates.

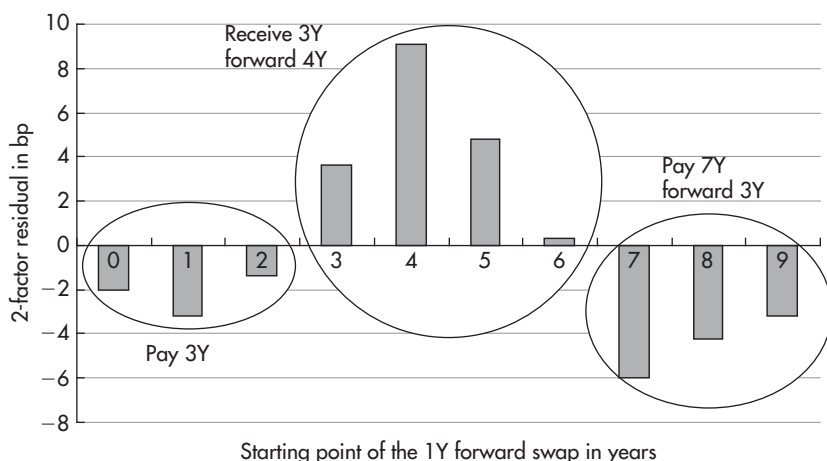


FIGURE 3.16 Two-factor residuals of a PCA on consecutive one-year forward EUR swap rates.

Sources: data – Bloomberg, authors; chart – Authors.